

Adaptive Kernel Selection for Stein Variational Gradient Descent

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Abstract

Stein Variational Gradient Descent (SVGD) is a popular variational inference method that transports a set of particles to approximate a target posterior, but its dynamics live in a reproducing kernel Hilbert space (RKHS) and are notoriously sensitive to the choice of kernel, which directly governs both convergence and approximation quality. The commonly used median heuristic sets kernel bandwidths cheaply but lacks flexibility and degrades markedly in high dimensions. This paper makes three contributions toward removing this bottleneck. First, we introduce Adaptive SVGD (Ad-SVGD), a method that alternates between transporting particles via SVGD and adaptively tuning kernel parameters by gradient ascent on the kernelized Stein discrepancy (KSD), over an abstract parameterized family of kernels rather than a single scalar bandwidth. Second, we provide a theoretical analysis that lifts existing fixed-kernel KSD convergence results to the adaptive setting, establishing that the maximal KSD over our kernel class converges to zero and, under a Stein log-Sobolev inequality, yielding mean-field iteration complexity. Third, we validate Ad-SVGD empirically and show that it consistently outperforms the median heuristic across a range of tasks while alleviating the variance collapse that plagues SVGD in high dimensions. Together, these results demonstrate that principled, adaptively chosen kernels alone can substantially extend the reach of SVGD.

1 Introduction

Stein Variational Gradient Descent (SVGD) [18] is a deterministic particle-based method for approximate Bayesian inference that has emerged as a popular alternative to traditional Markov Chain Monte Carlo (MCMC) methods. SVGD evolves a set of particles using update directions derived from the functional gradient of the Kullback-Leibler (KL) divergence to the target distribution, with updates constrained to lie within the unit ball of a reproducing kernel Hilbert space (RKHS). Its performance, however, hinges on a single design choice: the kernel. Kernel choice significantly influences the algorithm’s behavior [8, 28], and a poorly chosen kernel drives the particle approximation to underestimate the posterior variance [2]. These two failure modes have hardened into a widely held belief that SVGD simply does not scale, failing to perform well as the dimension of the underlying state space increases. The central message of this paper is that this belief is misdirected: much of the observed failure is a symptom of poor kernel selection, not of SVGD itself. We substantiate this message by introducing an adaptive mechanism that dynamically tunes the kernel during inference by maximizing the kernelized Stein discrepancy (KSD), enabling more effective transport in complex and high-dimensional spaces. In doing so, we isolate kernel selection as a lever that alone can recover competitive performance where the median heuristic collapses.

1.1 Related work.

Since its introduction [18], SVGD has become a widely used tool for approximate Bayesian inference across a range of machine learning applications [22, 24, 30, 15], and its theoretical foundations are by now well developed. Mean-field convergence has been analyzed in both continuous-time [23, 8, 5] and discrete-time [16, 32] settings, finite-particle convergence rates have been established under various assumptions [3, 33], and the dynamics have been connected to gradient flows on probability distribution spaces [19, 8] in analogy to Jordan, Kinderlehrer, and Otto [14]. Against this backdrop, our contribution is orthogonal: rather than proposing new dynamics, we ask how far the choice of kernel alone can carry the existing method.

The reason this question matters is that SVGD’s performance depends critically on the kernel, which determines both the interaction between particles and the overall convergence of the method (see also Figures 1 and 9). Convergence results typically concern mean-field convergence with respect to the KSD, whose relation to weak convergence itself depends on the selected kernel [11]. The commonly used median heuristic [12] is simple to implement but lacks theoretical justification and is known to degrade as dimensionality increases [2, 40, 36]. Prior efforts to counter this degradation fall into two camps: tools tailored to high-dimensional settings [7, 37, 10, 21], and modifications that use neural networks to learn the update direction [17, 39]. Both change the algorithm; our work instead keeps the algorithm fixed and adapts only the kernel.

Closest to our work is Ai et al. [1], which introduces a mixture-of-kernels framework: it defines a KSD for a weighted linear combination of kernels and learns the kernel weights by maximizing this multiple-kernel KSD. We share the principle of driving kernel selection by KSD maximization, but our contribution goes further. Their method is restricted to a finite kernel basis and fixed base kernels, whereas we optimize continuous kernel parameters over an abstract kernel family, which is what unlocks flexible, dimension-dependent bandwidths.

1.2 Contributions.

We frame our work around a single, deliberately narrow research question: *can SVGD, without any architectural or algorithmic enhancements, achieve competitive performance when equipped with a principled and adaptively chosen kernel?* By isolating kernel selection from every other modification found in SVGD variants, we can attribute any observed gains to kernel choice alone, and thereby measure how far adaptive kernel learning can push the capabilities of the original method.

Our answer to this question is affirmative, and it rests on three contributions.

- (i) **An adaptive kernel selection method (Section 3).** We propose Ad-SVGd, which dynamically updates the kernel parameters by maximizing the KSD during SVGD inference. In contrast to the median heuristic, which tunes a single scalar bandwidth, Ad-SVGd optimizes multiple continuous kernel parameters, giving the method the flexibility to adapt per-dimension bandwidths on the fly at no extra gradient cost.
- (ii) **A convergence analysis for the adaptive setting (Section 4).** We analyze Ad-SVGd in the discrete-time mean-field setting, lifting existing fixed-kernel convergence results to a kernel that changes at every step. Concretely, we prove that the supremum of the KSD over the parameterized kernel class converges to zero as the particle distribution approaches the target, and, under a Stein logarithmic Sobolev inequality, we further derive mean-field iteration-complexity bounds, requiring only a single kernel in the class to satisfy the inequality.

- (iii) **Empirical validation across dimensions (Section A and beyond).** Through numerical experiments spanning Gaussian mixtures, ODE-based inverse problems, and Bayesian logistic regression, we demonstrate that adaptive kernel selection consistently outperforms the median heuristic and, crucially, alleviates the variance collapse that worsens with dimension.

Taken together, these contributions support our central claim: kernel selection, done adaptively and by a principled criterion, is by itself enough to make plain SVGD competitive in regimes where it was thought to fail.

2 Mathematical background

Before presenting our method, we fix the notation and recall the two ingredients it builds on: the kernelized Stein discrepancy and the SVGD dynamics. We begin with a symmetric positive definite kernel $k : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ and its associated RKHS $\mathcal{H}_0$. We define $\mathcal{H}$ as the d -fold Cartesian product of $\mathcal{H}_0$ equipped with the inner product $\langle f, g \rangle_{\mathcal{H}} = \sum_{i=1}^d \langle f_i, g_i \rangle_{\mathcal{H}_0}$ and the canonical feature map $\Phi_k(x) = k(\cdot, x) \in \mathcal{H}_0$. Throughout, we denote by $x \cdot y$ the Euclidean inner product and $\nabla \cdot$ the divergence operator. The space of probability measures on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ is denoted by $\mathcal{P}(\mathbb{R}^d)$ and $\mathcal{P}_p(\mathbb{R}^d)$ denotes the subspace of measures with finite p -th moment. For $\mu, \nu \in \mathcal{P}_p(\mathbb{R}^d)$, we define the Wasserstein p -distance

$$\mathcal{W}_p(\mu, \nu) = \inf_{\gamma \in \Gamma(\mu, \nu)} \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|_2^p d\gamma(x, y) \right)^{1/p},$$

where $\Gamma(\mu, \nu)$ is the set of couplings of μ and ν , i.e. the set of probability measures on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ and ν .

2.1 Integral probability metrics and kernelized Stein discrepancy.

Integral probability metrics (IPMs) [26] quantify the distance between two measures as the maximum deviation of integrals over a class of test functions. To make this feasible for measuring the distance of a sample to an intractable target distribution, Stein’s method [35] constructs test functions with zero mean w.r.t. the target. Indeed, for the operator $\mathcal{S}_{\pi}f := \nabla \log \pi \cdot f + \nabla \cdot f$ and a suitable choice of kernel, we have $\int_{\mathbb{R}^d} \mathcal{S}_{\pi}f(x) d\pi(x) = 0$ for all $f \in \mathcal{H}$ [6, 20]. This gives rise to the *kernelized Stein discrepancy* (KSD), the criterion our adaptive method will later maximize over kernels,

$$\text{KSD}(\mu|\pi) := \sup_{f \in B(\mathcal{H})} \left| \int_{\mathbb{R}^d} \mathcal{S}_{\pi}f d\mu \right|, \quad (1)$$

where $B(\mathcal{H})$ denotes the unit ball in $\mathcal{H}$. This optimization problem admits the closed-form solution $f^* = \frac{\psi}{\|\psi\|_{\mathcal{H}}}$ with $\psi = \int_{\mathbb{R}^d} \mathcal{A}_{\pi}^k d\mu$, where $\mathcal{A}_{\pi}^k(x) = \nabla \log \pi(x) \Phi_k(x) + \nabla \Phi_k(x) \in \mathcal{H}$ and Φ_k is the feature map associated with the kernel k . Consequently, the supremum evaluates to the RKHS norm of ψ , giving the equivalent characterization $\text{KSD}(\mu|\pi) = \|\psi\|_{\mathcal{H}}$ that we use throughout.

2.2 Stein Variational Gradient Descent.

Given a target distribution π and reference distribution μ_0 , SVGD transforms μ_0 into an approximation of π by setting $\mu_{n+1} := T_{\sharp} \mu_n$, where $T_{\sharp}$ is the push-forward operator for the map

$T(x) = x + \gamma \psi^{\mu_n}(x)$ and the vector field $\psi^\mu = \int_{\mathbb{R}^d} \mathcal{A}_\pi^k d\mu$ for $\mu \in \mathcal{P}(\mathbb{R}^d)$ is the direction of steepest descent. The choice is not arbitrary: the solution of Equation (1) implies that $\frac{\psi^{\mu_n}}{\|\psi^{\mu_n}\|_{\mathcal{H}}}$ minimizes $\frac{d}{d\gamma} \text{KL}(T_\# \mu_n \parallel \pi) \Big|_{\gamma=0}$ over the unit ball of $\mathcal{H}$ [cf. 18, Theorem 3.1], where $\text{KL}(\cdot \parallel \cdot)$ denotes the KL-divergence. This tight link between the update direction and the KSD is precisely what our method exploits, since for this choice we have

$$\frac{d}{d\gamma} \text{KL}\left(\left(\text{Id} + \gamma \psi^{\mu_n}\right)_\# \mu_n \parallel \pi\right) \Big|_{\gamma=0} = -\text{KSD}^2(\mu_n | \pi). \quad (2)$$

Iteratively applying the maps T generated in this way to a particle set $\{X_0^i\}_{i=1}^M$ sampled from μ_0 yields the particle updates:

$$X_{n+1}^i = X_n^i + \frac{\gamma}{M} \sum_{j=1}^M k(X_n^i, X_n^j) \nabla \log \pi(X_n^j) + \nabla_{X_n^j} k(X_n^i, X_n^j). \quad (3)$$

3 Adaptive kernel selection for SVGD

This section presents our first contribution: Ad-SVGd, an adaptive kernel selection scheme for SVGD. We motivate the design criterion, state the resulting alternating algorithm, and account for its computational cost relative to vanilla SVGD.

We start from an observation about what SVGD convergence guarantees actually control. Because proving convergence of SVGD in KL divergence is challenging and requires restrictive assumptions, recent work has shifted attention to convergence in terms of the kernelized Stein discrepancy (KSD) [16, 33, 32]. Yet minimizing the KSD alone does not guarantee weak convergence: the sequence of mean-field measures may fail to be tight, a necessary condition for weak convergence of probability measures in Polish spaces. Moreover, as Equation (1) makes explicit, the RKHS structure — and hence the choice of kernel — directly determines how well convergence in KSD translates into weak convergence [11]. This is exactly the leverage point our method targets.

To turn this leverage into an algorithm, we formulate our adaptive variant of SVGD over a parameterized family of kernels $\{k_\theta \mid \theta \in \Theta\}$. For each $\theta \in \Theta$, we write KSD_θ for the corresponding *kernelized Stein discrepancy*, ψ_θ^μ for the optimal update direction, and Φ_θ for the associated feature map. This explicit parameterization is what lets us adaptively select kernels during optimization, so that the kernel’s influence on convergence is accounted for rather than fixed in advance.

The core design principle of Ad-SVGd is to select, at each step, the kernel that *maximizes* the KSD between the empirical particle distribution and the target. The rationale is direct: the instantaneous decrease in KL divergence under SVGD is proportional to the squared KSD at the current particle measure (see Equations (2) and (7)). Choosing the kernel with the largest KSD therefore maximizes the instantaneous rate of KL decrease. This principle is reinforced by geometric analyses of SVGD as a gradient flow, which show that kernels inducing larger KSD values yield more favorable convergence properties when comparing the associated RKHSs [28, 8].

Whereas vanilla SVGD guarantees convergence of the KSD for a single fixed kernel, Ad-SVGd instead targets the largest KSD attainable within the kernel class. Unlike the median heuristic of Liu and Wang [18], this principle applies to *any* class of parameterized kernels: at each step we seek an optimal parameter

$$\theta_n \in \operatorname{argmax}_{\theta \in \Theta} \text{KSD}_\theta(\mu_n | \pi).$$

Concretely, before each particle update we adjust the kernel parameter by one or more gradient ascent steps on the KSD with step size $s > 0$, and we optionally skip this adjustment on some iterations to save runtime (see Algorithm 1), governed by a user-specified decision variable `paramupdate`. Two modularity properties are worth highlighting, as they underline that our contribution is the kernel-selection mechanism rather than a specific solver: the base SVGD transport step can be swapped for more advanced variants (adaptive step-size schedules, line searches, momentum, or second-order/preconditioned updates) without touching kernel selection, and the gradient ascent on the kernel parameter can likewise be replaced by any alternative optimizer. Our implementation of the kernel update rests on the formula [6, 20]

$$\text{KSD}^2(\mu|\pi) = \int_{\mathbb{R}^d \times \mathbb{R}^d} u_\pi^k d(\mu \otimes \mu) \quad (4)$$

with

$$\begin{aligned} u_\pi^k(x, y) = & k(x, y) \nabla \log \pi(x) \cdot \nabla \log \pi(y) + \nabla \log \pi(y) \cdot \nabla_x k(x, y) \\ & + \nabla \log \pi(x) \cdot \nabla_y k(x, y) + \text{trace}(\nabla_x \nabla_y k(x, y)), \end{aligned} \quad (5)$$

from which the required gradient follows directly. A key selling point of Ad-SVGd is that this adaptivity is nearly free; two observations pin down its overhead relative to vanilla SVGD:

- The gradient ascent steps for the kernel parameter reuse the same gradients $\nabla \log \pi(X_n^i)$ already computed for the SVGD step, so Ad-SVGd requires *no* additional gradient evaluations of $\log \pi$ — typically the dominant cost.
- Any residual overhead can be trimmed further by using a small number of update steps `nstepstheta` for the kernel parameter, or by updating the kernel parameter only once every several SVGD steps. When the number of particles M is large, subsampled particles suffice to approximate the KSD empirically.

Algorithm 1: Ad-SVGD

Input: Initial particle set $\{X_0^i \in \mathbb{R}^d \mid i = 1, \dots, M\}$, kernel class $\{k_\theta \mid \theta \in \Theta\}$, initial kernel parameter $\theta_{-1} \in \Theta$, step sizes $\gamma, s > 0$, number of steps $nsteps, nstepstheta \in \mathbb{N}$

Output: Final particle set $\{X_{nsteps}^i \in \mathbb{R}^d \mid i = 1, \dots, M\}$

```
for  $n = 0$  to  $nsteps - 1$  do
    Decide  $paramupdate \in \{\text{True}, \text{False}\}$ ;
    if  $paramupdate$  then
         $\theta_n^0 \leftarrow \theta_{n-1}$ ;
        for  $\ell = 0$  to  $nstepstheta - 1$  do
             $\theta_n^{\ell+1} \leftarrow \theta_n^\ell + s \nabla_{\theta_n^\ell} \text{KSD}_{\theta_n^\ell}^2 \left( \frac{1}{M} \sum_{i=1}^M \delta_{X_n^i} \mid \pi \right)$ ;
        end
         $\theta_n \leftarrow \theta_n^{nstepstheta}$ ;
    else
         $\theta_n \leftarrow \theta_{n-1}$ ;
    end
    for  $i = 1$  to  $M$  do
         $X_{n+1}^i \leftarrow X_n^i + \frac{\gamma}{M} \sum_{j=1}^M k_{\theta_n}(X_n^i, X_n^j) \nabla \log \pi(X_n^j) + \nabla_{X_n^j} k_{\theta_n}(X_n^i, X_n^j)$ ;
    end
end
```

4 Convergence analysis

Our second contribution is a convergence theory for Ad-SVGD, and this section develops it. We consider a target measure with Lebesgue density $\pi(x) \propto \exp(-V)$ for a potential $V : \mathbb{R}^d \rightarrow \mathbb{R}$. The goal is to show that the adaptive scheme inherits, and in a precise sense strengthens, the convergence guarantees available for fixed-kernel SVGD. To this end we extend the analysis of [32], which operates under Talagrand’s inequality — a condition that holds under mild assumptions on the target and is weaker than the commonly employed logarithmic Sobolev inequality. We adopt the same assumptions as [32], now required to hold uniformly over all kernel parameters.

Assumption 1. *We assume that $V \in C^2$ such that $\int_{\mathbb{R}^d} \exp(-V(x)) dx < \infty$ and that the Hessian H of V is uniformly bounded w.r.t. the operator norm, i.e. there exists $L \geq 0$ such that $\|H(x)\|_{\text{op}} \leq L$ for all $x \in \mathbb{R}^d$.*

Assumption 2. *We assume that $\pi \in \mathcal{P}_1(\mathbb{R}^d)$ satisfies Talagrand’s inequality T1, which means that there exists $\lambda > 0$ such that*

$$\mathcal{W}_1(\mu, \pi) \leq \sqrt{2 \text{KL}(\mu \parallel \pi) / \lambda}$$

for all $\mu \in \mathcal{P}_1(\mathbb{R}^d)$.

Assumption 3. *We assume there exists $B > 0$ such that $\|\Phi_\theta(x)\|_{\mathcal{H}_0} \leq B$ for all $x \in \mathbb{R}^d$, $\theta \in \Theta$, and that $\nabla \Phi_\theta$ is continuous with $\|\nabla \Phi_\theta(x)\|_{\mathcal{H}}^2 = \sum_{i=1}^d \|\partial_i \Phi_\theta(x)\|_{\mathcal{H}_0}^2 \leq B^2$ for all $x \in \mathbb{R}^d$, $\theta \in \Theta$.*

4.1 Convergence analysis of SVGD under Talagrand's inequality

We first recall the fixed-kernel building block that our adaptive analysis extends. Writing

$$\mathcal{F}(\mu) := \text{KL}(\mu \parallel \pi).$$

we make use of the following fundamental inequality. Given a fixed kernel parameter $\theta \in \Theta$ satisfying Assumption 3, define the pushforward measure

$$\tilde{\mu} = (I + \gamma g) \# \mu$$

for arbitrary $g \in \mathcal{H}$. Under Assumptions 1 and 3, let $\gamma, B > 0$, $\alpha > 1$ and $g \in \mathcal{H}_\theta$ such that $\gamma \|g\|_{\mathcal{H}_\theta} \leq \frac{\alpha-1}{\alpha B}$. Then, according to Proposition 3.1 in [32], it holds

$$\mathcal{F}(\tilde{\mu}) \leq \mathcal{F}(\mu) + \gamma \langle \psi_\theta^\mu, g \rangle_{\mathcal{H}_\theta} + \frac{\gamma^2 K}{2} \|g\|_{\mathcal{H}_\theta} \quad (6)$$

with $K = (\alpha^2 + L)B$. For the iterates (3) with a fixed kernel parameter $\theta \in \Theta$ one can then derive the descent condition [32, Theorem 3.2],

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) \text{KSD}_\theta^2(\mu_n | \pi), \quad (7)$$

provided that

$$\gamma \leq (\alpha - 1) \left(\alpha B (1 + \|\nabla V(0)\|) + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1}. \quad (8)$$

The crux in establishing (7) is verifying $\gamma \|\psi_\theta^{\mu_n}\|_{\mathcal{H}_\theta} \leq \frac{\alpha-1}{\alpha B}$ via Talagrand's inequality (Assumption 2), which permits applying (6) with $g = \psi_\theta^{\mu_n}$. Condition (7) then yields

$$\lim_{n \rightarrow \infty} \text{KSD}_\theta^2(\mu_n | \pi) = 0,$$

since $\sum_{n=0}^{\infty} \text{KSD}_\theta^2(\mu_n | \pi) \leq c_\gamma^{-1} \mathcal{F}(\mu_0)$ for $c_\gamma = \gamma(1 - \frac{\gamma B(\alpha^2 + L)}{2}) > 0$. This is the fixed-kernel guarantee we now lift to the adaptive setting. The first step is to note that with an adaptive kernel parameter choice (θ_n) , the inequality (7) becomes

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) \text{KSD}_{\theta_n}^2(\mu_n | \pi).$$

4.2 Convergence analysis of Ad-SVGd.

We now state our main convergence results for Ad-SVGd, treating first the idealized case of exact KSD maximization and then the practically relevant case of approximate maximization. Suppose the adaptive SVGd iteration can be written in the simplified form

$$\begin{aligned} \mu_{n+1} &= (I + \gamma \psi_{\theta_n}^{\mu_n}) \# \mu_n, \\ \theta_n &\in \operatorname{argmax}_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n | \pi), \end{aligned} \quad (9)$$

meaning that

$$\|\psi_{\theta_n}^{\mu_n}\|_{\mathcal{H}_{\theta_n}} = \max_{\theta \in \Theta} \text{KSD}_{\theta}(\mu_n | \pi).$$

This exact formulation is available whenever the KSD maximization over the kernel parameter has a (unique) closed-form solution. That is precisely the situation in the multiple-kernel SVGD framework [1], which parameterizes the kernel as a convex combination of base kernels

$$k_{\theta}(x, y) = \sum_{i=1}^N \theta_i k_i(x, y) \quad \text{with} \quad \sum_{i=1}^N \theta_i = 1.$$

In the general setting the update θ_n in (9) must be approximated instead; there we assume $\arg\max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n | \pi) \neq \emptyset$ for all $n \in \mathbb{N}$.

Our first guarantee handles the exact case, and follows directly from Theorem 3.2 in [32]: the descent now bites on the *best* kernel in the class at every step.

Lemma 1. *Suppose that Assumptions 1-3 are satisfied. For any $\alpha > 1$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

there exists $c_{\gamma} > 0$ such that for all $n \in \mathbb{N}$

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - c_{\gamma} \max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n | \pi),$$

where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (9).

Corollary 2. *Under the same assumptions as in Lemma 1 it holds that $\lim_{n \rightarrow \infty} \max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n | \pi) = 0$.*

The exact case is instructive but idealized; in practice the inner KSD maximization is solved only approximately, and our theory must accommodate this. Suppose that $\max_{\theta \in \Theta} \text{KSD}_{\theta}(\mu | \pi) < \infty$ for any $\mu \in \mathcal{P}(\mathbb{R}^d)$ and consider the alternating scheme

$$\begin{aligned} \theta_n &= \Psi_n(\theta_{n-1}, \mu_n), \\ \mu_{n+1} &= (\text{Id} + \gamma \psi_{\theta_n}^{\mu_n})_{\#} \mu_n \end{aligned} \tag{10}$$

for some sequence of iterative update rules $\Psi_n : \Theta \times \mathcal{P}(\mathbb{R}^d) \rightarrow \Theta$ aimed at maximizing $\text{KSD}_{\theta}(\mu_n | \pi)$. In Algorithm 1, Ψ_n is the gradient ascent update for the KSD, though as noted above it could be any suitable iterative optimizer. The only property we ask of the update rule is the following asymptotic accuracy.

Assumption 4. *We assume that there exists a sequence $(\varepsilon_n)_{n \in \mathbb{N}}$ such that $\lim_{n \rightarrow \infty} \varepsilon_n = 0$ and*

$$\max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n | \pi) - \text{KSD}_{\theta_n}^2(\mu_n | \pi) \leq \varepsilon_n \quad \text{for all } n \in \mathbb{N}.$$

Under this assumption alone we obtain our main convergence guarantee for the approximate scheme.

Theorem 3. Suppose that Assumptions 1-3 are satisfied. Under Assumption 4, for any $\alpha > 1$ with

$$\gamma \leq (\alpha - 1) \left(\alpha B (1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}}) \right)^{-1},$$

it holds that $\lim_{n \rightarrow \infty} \max_{\theta \in \Theta} \text{KSD}_{\theta}(\mu_n | \pi) = 0$, where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (10).

Proof. Using Theorem 3.2 in [32] we obtain

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - c_{\gamma} \text{KSD}_{\theta_n}^2(\mu_n | \pi)$$

Since $(\mathcal{F}(\mu_n))_{n \in \mathbb{N}}$ is decreasing and bounded below by zero, it converges and, in particular, it is a Cauchy sequence. Hence, we have

$$c_{\gamma} \text{KSD}_{\theta_n}^2(\mu_n | \pi) \leq \mathcal{F}(\mu_n) - \mathcal{F}(\mu_{n+1}) \rightarrow 0$$

as $n \rightarrow \infty$. Under Assumption 4 we obtain

$$\max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n | \pi) \leq \text{KSD}_{\theta_n}^2(\mu_n | \pi) + \varepsilon_n \rightarrow 0$$

as $n \rightarrow \infty$. □

This result can be strengthened. Under the so-called Stein logarithmic Sobolev inequality, we upgrade the convergence of Ad-SVGD from the KSD to the KL divergence itself, and for specific decay rates of ε_n we obtain explicit iteration-complexity bounds for the mean-field limiting system (10). This refined analysis also pins down the bias of Ad-SVGD when the KSD maximization is solved only to a fixed $\bar{\varepsilon}$ -accuracy per iteration; we defer the details to Section B. We stress one feature of the adaptive setting that is genuinely new here: we require only *one* kernel in the class $\{k_{\theta}\}$ to satisfy the Stein logarithmic Sobolev inequality, which makes the condition far easier to verify than in the fixed-kernel case.

5 Numerical Experiments

This section presents our third contribution: empirical evidence that adaptive kernel selection outperforms the median heuristic. We report two experiments here — a one-dimensional Gaussian mixture model and a linear inverse problem governed by an ordinary differential equation — chosen to expose, respectively, SVGD’s sensitivity to bandwidth and its tendency to underestimate posterior uncertainty. Section A adds toy experiments that isolate the effect of increasing dimensionality, and Section 5.4 demonstrates the advantages of Ad-SVGD over the median heuristic on a Bayesian logistic regression model drawn from prior SVGD literature. Across all of these, one message recurs: Ad-SVGD matches or beats the median heuristic and, unlike it, does not collapse as the dimension grows.

5.1 Kernel parameterization

To make the comparison concrete, we fix the kernel family used throughout the experiments. SVGD is most commonly run with kernels of the form

$$k_h(x, y) = \exp \left(-\frac{\|x - y\|_p^p}{h} \right),$$

where $\|\cdot\|_p$ denotes the p -norm on $\mathbb{R}^d$ [e.g. 18, 2, 8], and the object of selection is the scalar *bandwidth* h . The baseline we compare against, the median heuristic, sets $h = \frac{\text{med}^p}{\log(M-1)}$, where med is the current median distance between particles, a choice motivated by balancing the two terms in the SVGD update (3) [18].

The advantage Ad-SVGd brings is that it need not settle for a single scalar. We use product kernels

$$k_h(x, y) = \prod_{i=1}^d \exp\left(-\frac{|x_i - y_i|^p}{h_i}\right)$$

with parameter $h = (h_1, \dots, h_d)$, i.e. a separate bandwidth per dimension; the derivatives needed to apply Algorithm 1 to these kernels (see Equations (4) and (5)) are elementary. For a fair comparison we also tried extending the median heuristic to these kernels by taking a naive per-dimension median, but this variant performed poorly and suffered variance collapse — reinforcing that it is the KSD-driven adaptation, not merely the per-dimension parameterization, that carries the gains.

We note that the same flexibility opens the door to more scalable kernel families beyond those we test here: spectral or low-rank parametrizations can track the dominant directions of variability without the exponential cost of full product kernels, while Matérn or mixed-product constructions offer additional control over smoothness and anisotropy. These are natural extensions of the mechanism we contribute rather than part of our present evaluation.

5.2 Toy example

This first example establishes the problem our method solves: SVGD is acutely sensitive to the bandwidth. We take the one-dimensional two-component Gaussian mixture from Liu and Wang [18], $\pi = \frac{1}{3}\mathcal{N}(-2, 1) + \frac{2}{3}\mathcal{N}(2, 1)$, using $M \in \{50, 200, 500\}$ particles initialized from $\mathcal{N}(0, 1)$. We run SVGD for 10^4 steps at step size 1, with kernels of the form above ($p = 1$) and various *fixed* bandwidths h . Sample quality is measured by the Wasserstein 1-distance $\mathcal{W}_1$, computed via the explicit formula $\mathcal{W}_1(\mu, \nu) = \int_{\mathbb{R}} |F_\mu(x) - F_\nu(x)| dx$ (see Panaretos and Zemel [29]) against an exact sample of size 10^5 . As Figure 1 shows, the algorithm performs well only for bandwidths in a narrow range and degrades sharply outside it. This acute sensitivity is exactly why a principled, automatic selection strategy — our contribution — is needed rather than a fixed or heuristic choice.

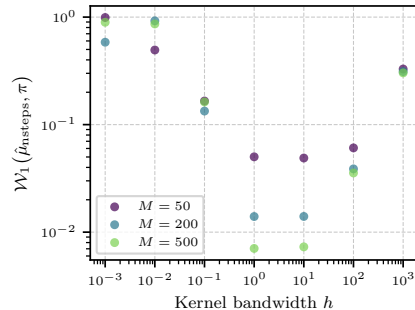


Figure 1: SVGD is highly sensitive to the bandwidth: final Wasserstein 1-distances for the one-dimensional example under fixed bandwidths h , with good performance confined to a narrow range of h . This acute sensitivity is exactly why a principled, automatic selection strategy — our contribution — is needed rather than a fixed or heuristic choice.

5.3 Linear inverse problem based on ODE

Having shown that bandwidth matters, we now demonstrate the payoff of adapting it: accurate posterior uncertainty on a Bayesian inverse problem where the median heuristic collapses. The

example is adapted from [38, Example 2.1]. We consider the one-dimensional differential equation

$$\begin{cases} -f''(s) + f(s) = u(s) & \text{for } s \in (0, 1) \\ f(s) = 0 & \text{for } s \in \{0, 1\} \end{cases} \quad (11)$$

and the associated inverse problem of recovering the right-hand side $u(\cdot) \in L^2([0, 1])$ from discrete noisy observation points of the solution $f \in H^2([0, 1]) \cap H_0^1([0, 1])$. These observations are described by

$$y = \Phi(u) + \varepsilon \in \mathbb{R}^{N_{\text{obs}}},$$

where $\varepsilon \in \mathbb{R}^{N_{\text{obs}}}$ is observational noise and the forward operator $\Phi : L^2([0, 1]) \rightarrow \mathbb{R}^{N_{\text{obs}}}$ is defined by $\mathcal{O} \circ H^{-1}$, with $H(f) = -f'' + f$ and $\mathcal{O}(f) = (f(s_1), \dots, f(s_{N_{\text{obs}}}))^\top \in \mathbb{R}^{N_{\text{obs}}}$ being the observation operator at N_{obs} equidistant points $s_k = \frac{k}{N_{\text{obs}}}$, $k = 1, \dots, N_{\text{obs}}$.

For the Bayesian formulation of the inverse problem, we consider a Gaussian process (GP) prior for u given by the truncated Karhunen-Loève (KL) expansion

$$u(\cdot, x) = Ax = \sum_{k=1}^{N_x} x_k \psi_k,$$

where $\psi_k(s) = \sqrt{2} \sin(\pi k s)$ and $x_k \sim \mathcal{N}(0, \lambda_k)$ independently with $\lambda_k = 50k^{-2}$. The resulting problem is to estimate the KL coefficients $x = (x_1, \dots, x_{N_x})^\top \in \mathbb{R}^{N_x}$ with prior given by $\mathcal{N}(0, \Gamma_0)$ with $\Gamma_0 = \text{diag}(\lambda_1, \dots, \lambda_{N_x})$. Assuming $\varepsilon \sim \mathcal{N}(0, \Gamma)$, this leads to the posterior density

$$\pi(x) \propto \exp \left(-\frac{1}{2} \|\Gamma^{-1/2}(y - \Phi Ax)\|^2 - \frac{1}{2} \|\Gamma_0^{-1/2} x\|^2 \right), \quad x \in \mathbb{R}^{N_x}.$$

For the implementation, we replace H by a numerical discretization operator for Equation (11) using a grid with mesh size 2^{-8} and consider the fully observed system (i.e. $N_{\text{obs}} = 2^8$). We use $N_x = 16$ terms for the KL expansion of u and assume noise covariance $\Gamma = 10^{-3} \text{Id}_{N_{\text{obs}}}$. We construct reference observations $\bar{y} \in \mathbb{R}^{N_{\text{obs}}}$ by drawing $\bar{x} \sim \mathcal{N}(0, \Gamma_0)$ and setting $\bar{y} = \Phi A \bar{x}$.

We compare SVGD with the median heuristic (Med-SVG) against Ad-SVG for particle ensemble sizes $M \in \{50, 100, 200\}$, using the kernels of Section 5.1 with $p = 1$. Both algorithms run for $4 \cdot 10^5$ iterations at particle step size 10^{-3} with a variant of AdaGrad for adaptive step-size control, as suggested by Liu and Wang [18]. For Ad-SVG we use step size 10^{-5} for the bandwidth updates and refresh the bandwidth only once every 100 particle updates; under this schedule the two methods showed no significant runtime difference, so any accuracy gain comes essentially for free.

The results deliver our central experimental finding. Figure 2 shows the GP reconstruction for an exemplary seed: both methods recover the mean well, but only Ad-SVG captures the posterior uncertainty. To quantify this, we use the Wasserstein 2-distance $\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)$ between the posterior π and the fitted normal $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$, with $\hat{\mu}$ the sample mean and $\hat{\Sigma}$ the sample covariance of the particle set [29, Section 3.2]. Since π is multivariate normal, this admits the explicit formula

$$\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)^2 = \|\hat{\mu} - \mu_\pi\|^2 + \text{trace} \left(\hat{\Sigma} + \Sigma_\pi - 2(\hat{\Sigma}^{1/2} \Sigma_\pi \hat{\Sigma}^{1/2})^{1/2} \right),$$

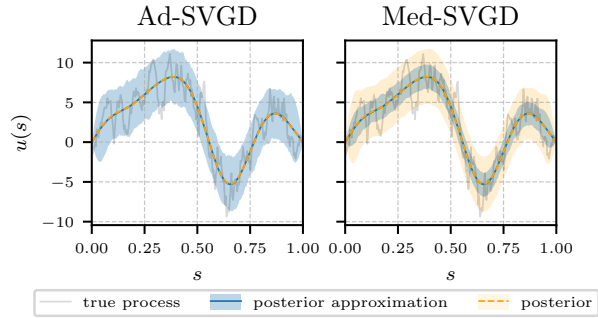
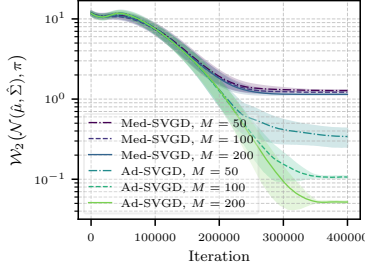
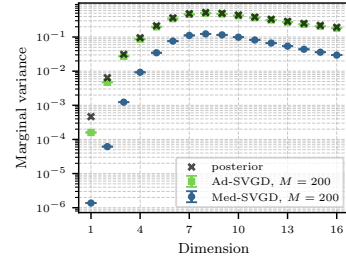


Figure 2: GP reconstruction for the ODE-based inverse problem (mean and 90% confidence interval): both methods recover the mean, but only Ad-SVG captures the posterior uncertainty.

where μ_π and Σ_π are the posterior mean and covariance. We additionally compare the marginal variances of the final particle distribution with those of the posterior. Aggregated over 56 random seeds (the posterior covariance does not depend on $\bar{y}$), the results in Figure 3 confirm the pattern: Ad-SVGD approximates the posterior better than Med-SVGD, which underestimates its uncertainty, and — unlike Med-SVGD — Ad-SVGD keeps improving as the particle count grows beyond 50. Figure 4 inspects the bandwidths Ad-SVGD discovers. The component-wise bandwidths stabilize faster than the approximation error and differ clearly across components, with the final bandwidths negatively correlated with the corresponding marginal variances — direct evidence that the adaptive mechanism is exploiting per-dimension structure the median heuristic cannot.

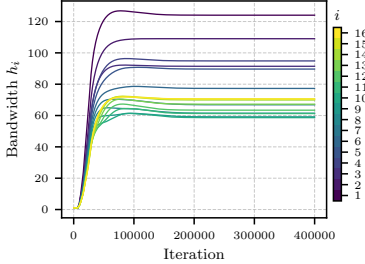


(a) Convergence of approximation error

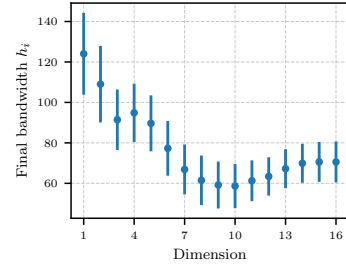


(b) Marginal variances

Figure 3: ODE-based inverse problem, Med-SVGD vs. Ad-SVGD (mean and 95% confidence interval over 56 random seeds): Ad-SVGD attains lower approximation error and recovers the marginal variances that Med-SVGD underestimates.



(a) Evolution of bandwidth per dimension (mean)



(b) Final bandwidths (mean and 95% conf. interval)

Figure 4: Bandwidths learned by Ad-SVGD on the ODE-based inverse problem (aggregated over 56 random seeds): component-wise bandwidths stabilize quickly and differ across dimensions.

5.4 Bayesian logistic regression

To show the gains carry over to a standard benchmark from the SVGD literature, we turn to a Bayesian logistic regression (BLR) model on the Covertypes data set [4], following Liu and Wang [18] and Liu et al. [21]. For the regression weights w we place a Gaussian prior $w \mid \alpha \sim \mathcal{N}(0, \alpha^{-1})$ with $\alpha \sim \text{Gamma}(1, 0.01)$ and infer the posterior of $x = [w, \alpha]$. Following Liu et al. [21], we generate a reference with the No-U-Turn Sampler (NUTS) of Hoffman and Gel-

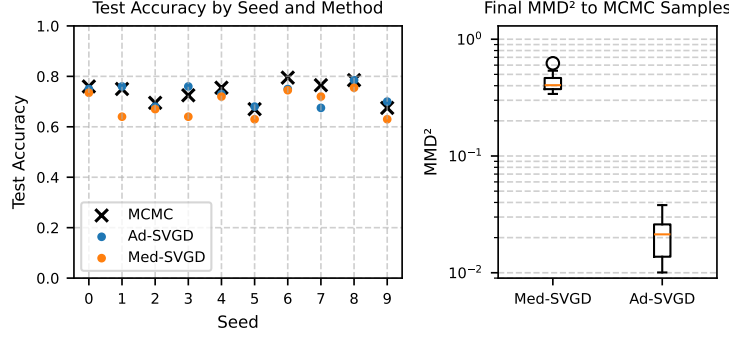


Figure 5: Bayesian logistic regression: Ad-SVGD improves on Med-SVGD in both prediction accuracy (left) and MMD^2 to the MCMC reference samples (right).

man [13] against which we assess the posterior approximation quality of Med-SVGD and Ad-SVGD. Again following Liu et al. [21], we test on size-1000 subsets of the data set and aggregate over 10 random draws, reporting classification performance as the test-set prediction accuracy of the particle mean. Figure 5 shows that Ad-SVGD outperforms Med-SVGD across almost all seeds, delivering both higher test accuracy and closer agreement with the MCMC reference.

To gauge how faithfully the posterior itself is captured, the same figure reports the squared maximum mean discrepancy (MMD^2) [25, Section 3.5] to the MCMC samples, which reveals a pronounced gap between the two methods. Figure 7 confirms the source of this gap: comparing the final particle covariance matrices to the MCMC reference (averaged over the 10 seeds, in the visualization style of Liu et al. [21]), Med-SVGD severely underestimates the posterior uncertainty while Ad-SVGD tracks it closely — the same variance-collapse story seen in the earlier experiments. Finally, Figure 6 shows that, starting from an uninformed initialization (bandwidth 1 in every component), Ad-SVGD recovers component-wise bandwidths that differ from the initialization by several orders of magnitude, underscoring how far an adaptive scheme must travel from any fixed default.

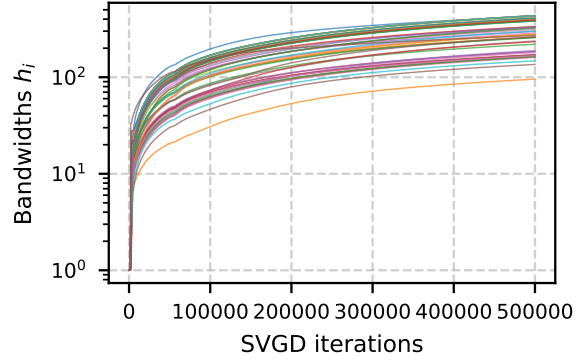


Figure 6: Bandwidth evolution for the BLR example: Ad-SVGD moves each component’s bandwidth by orders of magnitude from its uninformed initialization.

6 Limitations and outlook

We close by delimiting what our contributions do and do not establish. The main limitation concerns the second contribution, our theory: it rests on Assumption 4, which we do not prove holds for the alternating gradient-ascent scheme of Algorithm 1 used in our experiments. Establishing when this assumption is met is the subject of ongoing work; for now, the strong empirical results

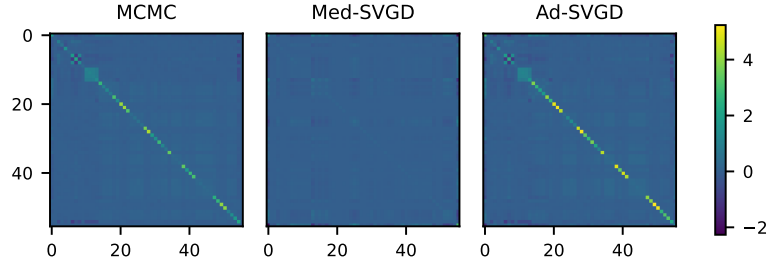


Figure 7: Covariance matrices of the final SVGD particle sets against MCMC samples (mean over 10 seeds): Med-SVG collapses the posterior variance, whereas Ad-SVG reproduces it.

stand as evidence that the gap between assumption and practice is benign.

A second boundary is one of scope rather than validity. Our study deliberately isolates kernel selection within the *original* SVGD dynamics, which is what lets us attribute the observed gains to kernel choice alone. A natural and promising next step, complementary to our contribution, is to combine adaptive kernel selection with recent SVGD variants such as sliced SVGD [10], Grassmann SVGD [21], or Stein transport [27], since our mechanism is modular by design and does not compete with these dynamics.

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A More numerical experiments

A.1 Gaussian mixture models

This appendix supplies additional evidence for our third contribution, tracing how the advantage of Ad-SVGD grows with dimension. We revisit the example from Section 5.2. For that setup, Figure 9 shows histograms of the final particle distribution against the target density for $h \in \{0.001, 1, 1000\}$, again illustrating the bandwidth sensitivity that motivates the method. We then compare Med-SVGD with Ad-SVGD across particle counts M , examining the final particle distributions after 10^4 iterations at step size 1.

Figure 8 reports the Wasserstein 1-distance between the final empirical particle distribution and the target for Med-SVGD and Ad-SVGD with $M = 10, 20, 50, 100, 200, 500$. As expected, accuracy improves with N for both methods, and the two are essentially on par here, each reaching a Wasserstein distance below 0.01 at $N = 500$. This parity is the point: in one dimension there is little for adaptivity to exploit, so the gains we report elsewhere are genuinely a high-dimensional phenomenon.

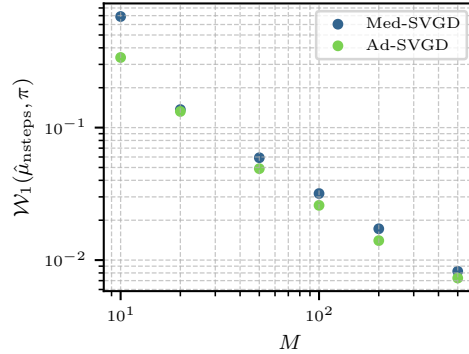


Figure 8: Approximation error for the 1D Gaussian mixture: Med-SVGD and Ad-SVGD perform comparably in one dimension.

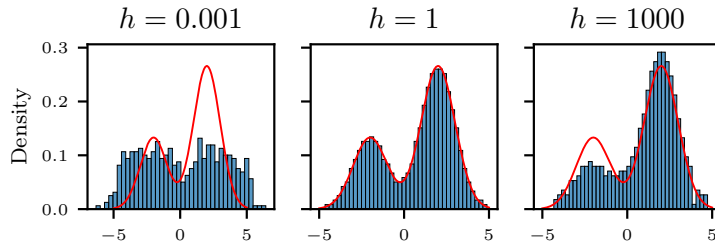


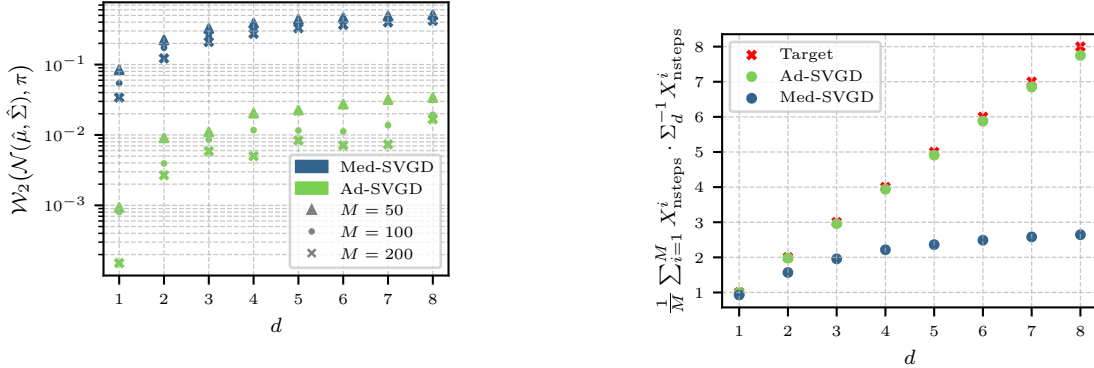
Figure 9: Final particle histograms under fixed bandwidths $h = 0.001, 1, 1000$, with target density π in red: the fit deteriorates badly away from a well-chosen h .

To summarize the one-dimensional case: both methods approximate the target well, and while our adaptive bandwidth selection performs on par, it holds no significant advantage over the median heuristic here. The next experiments show where that advantage does emerge.

A.2 Scaling dimension: multivariate normal distribution

This experiment is where the central claim of the paper is put to the test directly. We consider Gaussian targets $\pi_d = \mathcal{N}(0, \Sigma_d)$ with $\Sigma_d = \text{diag}(1, \frac{1}{2}, \dots, \frac{1}{d})$ for $d \in \{2, \dots, 8\}$, initialize particles from $\mathcal{N}(0, \frac{1}{d} \otimes^d)$, and run for 10^4 iterations at step size 0.1 (reducing the step size and lengthening the run where numerical stability demands it).

We track approximation quality as d grows. Since the Wasserstein 2-distance between two Gaussians has an explicit formula (see [29]), we measure sample quality by the Wasserstein 2-distance between π_d and $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$, with $\hat{\mu}$ and $\hat{\Sigma}$ the empirical mean and covariance of the particles [29, Section 3.2]; following the Python Optimal Transport library [9] used in our calculations, we call this the *Bures Wasserstein distance*. Figure 10a traces this measure for Med-SVGD and Ad-SVGD with $M = 50, 100, 200$ against dimension. The result is unambiguous and is the strongest evidence for our thesis: Ad-SVGD significantly outperforms Med-SVGD at every value of d , confirming that adaptive kernel selection alone lets plain SVGD scale where the median heuristic does not.



(a) Approximation error vs. dimension for $M \in \{50, 100, 200\}$: Ad-SVGD beats Med-SVGD at every d .

(b) χ^2 -test statistic for $M = 200$: Ad-SVGD stays near the true value d , Med-SVGD drifts away.

Figure 10: Multivariate normal targets of increasing dimension: Ad-SVGD maintains accuracy and correct scale as d grows, while Med-SVGD degrades.

Because the Bures Wasserstein distance is relatively insensitive to the small variances in the last components of π_d , we add a second, complementary diagnostic that targets exactly those components. The statistic $X \cdot \Sigma_d^{-1} X$ is χ^2 -distributed with d degrees of freedom — hence has expected value d — for $X \sim \mathcal{N}(0, \Sigma_d)$ (cf. [34, p. 27]). We compute its mean over the particles with $M = 200$ and compare it to d in Figure 10b. The two methods separate cleanly: Med-SVGD drifts ever further below the true expectation as d increases, whereas Ad-SVGD stays close to it. This is the variance-collapse failure of Med-SVGD made quantitative, and it is precisely the failure Ad-SVGD avoids.

Tables 1 and 2 drill down to the individual marginal variances for $M = 200$ across d , with the target variances listed in the first row of each table (Med-SVGD in Table 1, Ad-SVGD in Table 2). The contrast is stark and component-by-component: Ad-SVGD matches the target variances in every component, while Med-SVGD substantially underestimates them — consistent with Figure 10b.

Finally, we zoom in on the marginal distributions for $M = 200$ and $d = 8$. For clarity we normalize each component by the inverse of the corresponding target marginal standard deviation (i.e. we multiply the i -th component by i), which turns every marginal of π_d into a standard normal. Figures 11 and 12 show the resulting normalized marginal histograms for Med-SVGD and Ad-SVGD, with a standard Gaussian density overlaid. Once more, Ad-SVGD captures all marginals well while Med-SVGD underestimates the uncertainty; the Q-Q plots in Figures 13 and 14, comparing the normalized marginal quantiles against a standard normal, tell the same story from another angle.

Table 1: Med-SVGD ($M = 200$) marginal variances of the final particle distribution for the d -dimensional examples, with target variances of π_d in the first row: note the systematic underestimation across components.

component	1	2	3	4	5	6	7	8
Target	1.0000	0.2500	0.1111	0.0625	0.0400	0.0278	0.0204	0.0156
$d = 1$	0.9285							
$d = 2$	0.7921	0.1943						
$d = 3$	0.6803	0.1625	0.0697					
$d = 4$	0.6089	0.1440	0.0593	0.0311				
$d = 5$	0.5532	0.1275	0.0526	0.0271	0.0157			
$d = 6$	0.5190	0.1190	0.0481	0.0243	0.0140	0.0089		
$d = 7$	0.4900	0.1122	0.0449	0.0228	0.0131	0.0081	0.0052	
$d = 8$	0.4753	0.1077	0.0430	0.0215	0.0122	0.0074	0.0047	0.0032

Table 2: Ad-SVGD ($M = 200$) marginal variances of the final particle distribution for the d -dimensional examples, with target variances of π_d in the first row: the target is matched closely in every component.

component	1	2	3	4	5	6	7	8
Target	1.0000	0.2500	0.1111	0.0625	0.0400	0.0278	0.0204	0.0156
$d = 1$	0.9953							
$d = 2$	0.9907	0.2472						
$d = 3$	0.9867	0.2459	0.1095					
$d = 4$	0.9881	0.2467	0.1095	0.0610				
$d = 5$	0.9840	0.2433	0.1096	0.0616	0.0392			
$d = 6$	0.9858	0.2459	0.1090	0.0611	0.0392	0.0269		
$d = 7$	0.9856	0.2463	0.1086	0.0613	0.0390	0.0269	0.0199	
$d = 8$	0.9691	0.2409	0.1085	0.0611	0.0390	0.0268	0.0196	0.0150

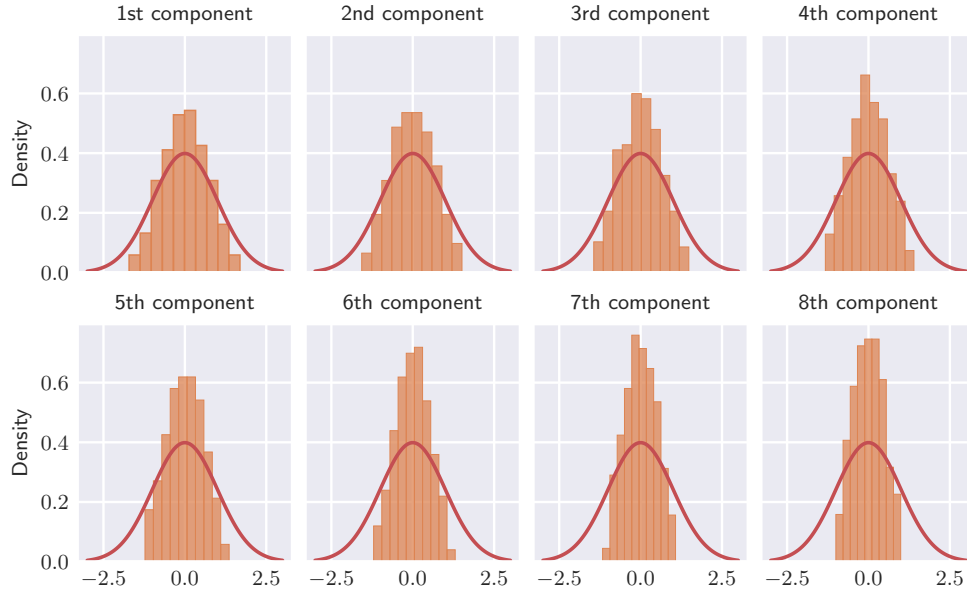


Figure 11: Med-SVGd ($M = 200$), eight-dimensional example: histograms of the final particle components, rescaled by the marginal target variances, with a standard Gaussian overlaid — the marginals are visibly too concentrated.

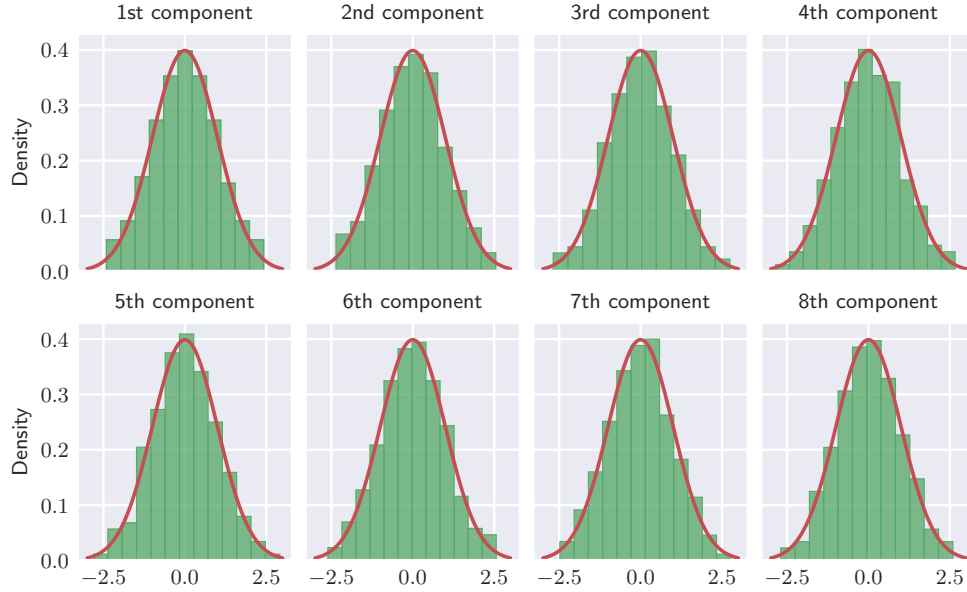


Figure 12: Ad-SVGd ($M = 200$), eight-dimensional example: histograms of the final particle components, rescaled by the marginal target variances, with a standard Gaussian overlaid — the marginals closely match the target.

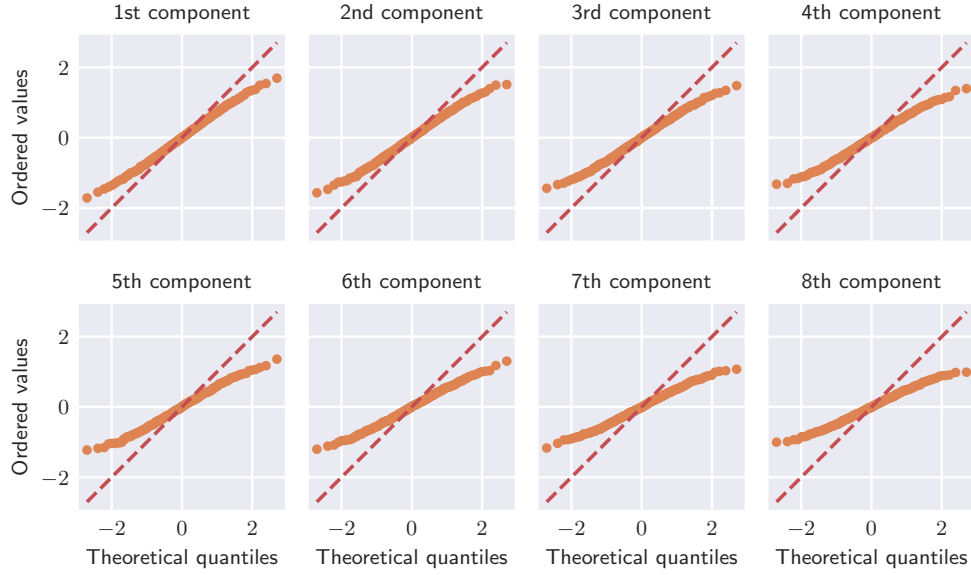


Figure 13: Med-SVGd ($M = 200$), eight-dimensional example: Q-Q plots of the rescaled marginal particle distributions against a standard normal, with the identity line for reference — the deviation signals underdispersed marginals.

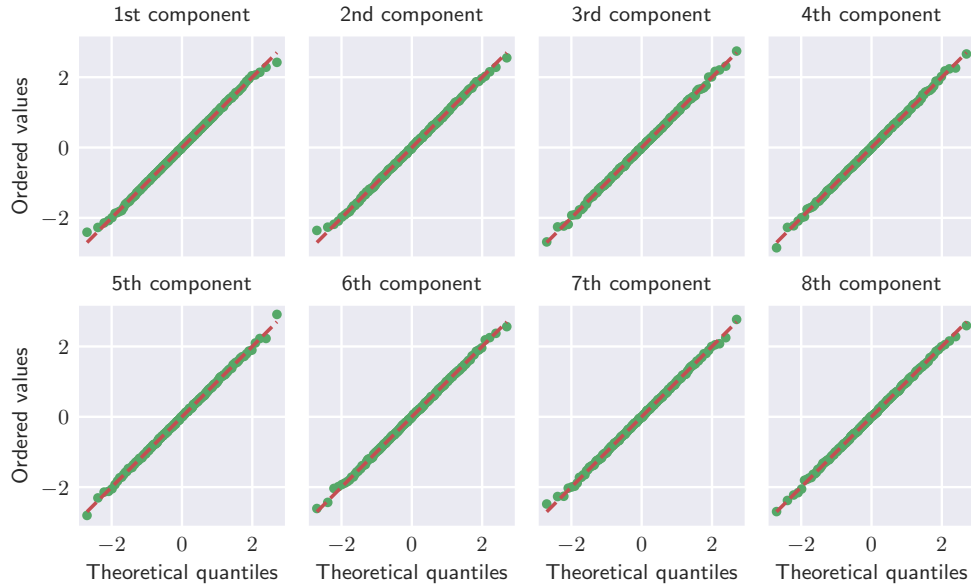


Figure 14: Ad-SVGd ($M = 200$), eight-dimensional example: Q-Q plots of the rescaled marginal particle distributions against a standard normal, with the identity line for reference — the marginals align with the target.

A.3 Gaussian process inference

As a final stress test of our third contribution under scaling dimension, we study a Gaussian process (GP) inference task from [31]. We consider a GP on $[0, 1]$ represented by a truncated Karhunen-Loève (KL) expansion $u(s, x) = \sum_{k=1}^{N_x} x_k \psi_k(s)$ with basis functions $\psi_k(s) = \sqrt{2} \sin(k\pi s)$, where $x = (x_1, \dots, x_{N_x})^\top$ is a vector of independent Gaussian random variables $x_k \sim \mathcal{N}(0, k^{-2})$. We observe the process at N_y equispaced points in $[0, 1]$ and infer the coefficients x_k . For fixed N_x and N_y , this corresponds to an inverse problem with forward model $Y = AX + \varepsilon$, where the k -th column of $A \in \mathbb{R}^{N_y \times N_x}$ is $(\psi_k(s_1), \dots, \psi_k(s_{N_y}))^\top$, $s_i = \frac{i}{N_y}$ for $i = 1, \dots, N_y$. The prior is $X \sim \mathcal{N}(0, \Sigma)$ with diagonal matrix Σ with entries k^{-2} , $k = 1, \dots, N_x$ and we assume independent Gaussian noise $\varepsilon \sim \mathcal{N}(0, I_{N_y})$. We construct reference observations $\bar{y} \in \mathbb{R}^{N_y}$ by drawing $\bar{x} \sim \mathcal{N}(0, \Sigma)$ and setting $\bar{y} = A\bar{x}$.

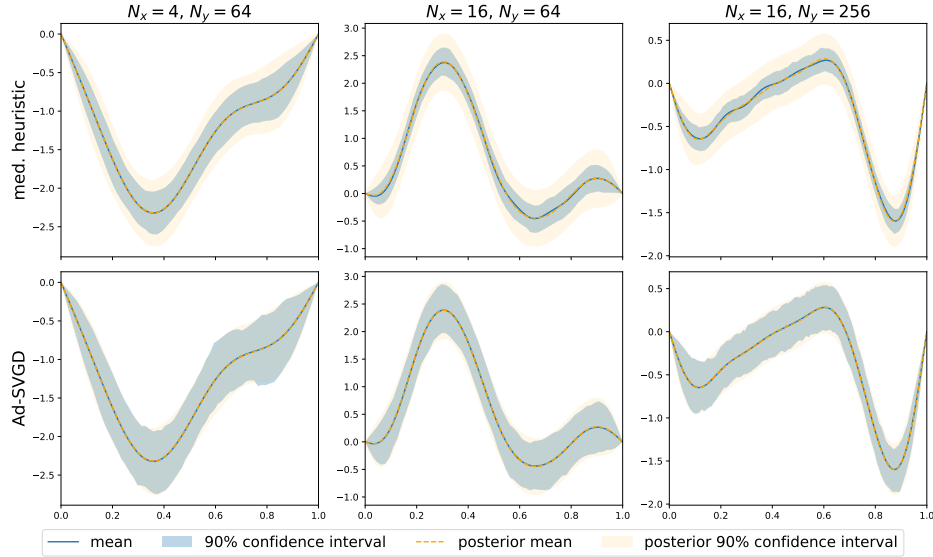


Figure 15: Estimated processes from the median heuristic and Ad-SVGD ($M = 100$ particles) across model configurations, compared to the posterior: Ad-SVGD tracks the posterior spread as the dimension grows, the median heuristic does not.

We use SVGD to sample from the resulting posterior $X|Y = \bar{y}$ and again pit the median heuristic against our adaptive approach. The median heuristic uses kernels $k(x, y) = \exp(-\|x - y\|_1/h)$, while Ad-SVGD uses product kernels $k(x, y) = \prod_{i=1}^{N_x} \exp(-|x_i - y_i|/h_i)$ with a separate bandwidth h_i per dimension; following [18], we use a variant of Adagrad for step-size control in the $N_x = 16$ scenarios.

Figure 15 compares the estimated processes to the posterior for $(N_x, N_y) \in \{(4, 64), (16, 64), (16, 256)\}$. The now-familiar pattern reappears: as the dimension increases the median heuristic underestimates the posterior variance while Ad-SVGD stays accurate, and this holds consistently across the number of GP observations. Table 3 makes the comparison quantitative through the trace of the particle covariance against the true posterior — the median heuristic severely undershoots the uncertainty, whereas Ad-SVGD recovers it closely. This closes the empirical case for our third contribution: across every scaling experiment, adaptive kernel selection is what keeps SVGD faithful in high dimensions.

N_x	4	8	16	16	16
N_y	64	64	64	128	256
theoretical	0.056	0.083	0.086	0.051	0.029
med. heuristic	0.026	0.023	0.022	0.012	0.006
Ad-SVGD	0.055	0.072	0.074	0.044	0.026

Table 3: Trace of the final particle covariance ($M = 100$ particles) against the theoretical posterior across configurations, averaged over 25 runs: Ad-SVGD stays close to the theoretical trace while the median heuristic collapses it.

B Refined theoretical analysis

This appendix elaborates the strengthened form of our second contribution announced in Section 4. Under a so-called Stein log-Sobolev inequality, which relates the KL divergence to the KSD, the convergence guarantee sharpens into an explicit rate. Duncan, Nüsken, and Szpruch [8] introduced this inequality and discussed when it holds. The adaptive setting buys us a genuine relaxation that the fixed-kernel setting does not enjoy: we require only *one* kernel in the entire class to satisfy the log-Sobolev inequality. We formalize this as follows.

Assumption 5. *Given a kernel class $\{k_\theta \mid \theta \in \Theta\}$, we assume that $\pi \in \mathcal{P}_1(\mathbb{R}^d)$ satisfies the generalized Stein log-Sobolev inequality with constant $\lambda > 0$:*

$$2\lambda \text{KL}(\mu \parallel \pi) \leq \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu \parallel \pi)$$

for all $\mu \in \mathcal{P}_1(\mathbb{R}^d)$.

With this property in hand we quantify convergence directly in the KL divergence. The practical strength of the result is that it bounds $\mathcal{F}(\mu_n)$ even under the weaker approximate condition $\varepsilon_n \leq \bar{\varepsilon}$, i.e. without requiring $\varepsilon_n \rightarrow 0$ — so a persistently inexact inner maximization still yields a controlled bias.

Theorem 4. *Suppose that Assumptions 1 to 3 and 5 are satisfied. For any $\alpha > 1$ and $\gamma > 0$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

and

$$\rho := 1 - 2\lambda c_\gamma \in (0, 1) \quad \text{where} \quad c_\gamma := \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2} \right),$$

it holds that

$$\mathcal{F}(\mu_{n+1}) \leq \rho \mathcal{F}(\mu_n) + c_\gamma \varepsilon_n,$$

where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (10) and $\varepsilon_n \geq \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n \parallel \pi) - \text{KSD}_{\theta_n}^2(\mu_n \parallel \pi)$. In particular, if

1. $\varepsilon_n \leq \bar{\varepsilon}$ for all $n \in \mathbb{N}$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + \frac{c_\gamma \bar{\varepsilon}}{1 - \rho}$$

for all $n \in \mathbb{N}$.

2. $\varepsilon \leq c_\varepsilon q^n$ for some $q \in (0, 1)$, $c_\varepsilon > 0$ and all $n \in \mathbb{N}$. Under $\rho \neq q$ we have

$$\mathcal{F}(\mu_n) \in \mathcal{O}(\max\{\rho, q\}^n)$$

and under $\rho = q$ we have

$$\mathcal{F}(\mu_n) \in \mathcal{O}(\tilde{q}^n)$$

for all $\rho < \tilde{q} < 1$.

3. $\varepsilon_n \leq \frac{c_\varepsilon}{(n+1)^p}$ for $c_\varepsilon, p > 0$ and all $n \in \mathbb{N}$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + \frac{2^p c_\gamma c_\varepsilon}{n^p} \frac{1}{1 - \rho} + c_\gamma c_\varepsilon \frac{\rho^{\lfloor n/2 \rfloor}}{1 - \rho}$$

for all $n \in \mathbb{N}$.

Proof. Similar to the proof of Theorem 3, under Assumption 5 and the conditions on α, γ , we have

$$\begin{aligned} \mathcal{F}(\mu_{n+1}) &\leq \mathcal{F}(\mu_n) - c_\gamma \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n \mid \pi) + c_\gamma \varepsilon_n \\ &\leq \rho \mathcal{F}(\mu_n) + c_\gamma \varepsilon_n, \end{aligned}$$

where $c_\gamma = \gamma(1 - \frac{\gamma B(\alpha^2 + L)}{2}) > 0$. Iterating this recursion over $n \in \mathbb{N}$, we deduce the general error bound

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + c_\gamma \sum_{k=0}^{n-1} \rho^{n-1-k} \varepsilon_k.$$

We consider the three different cases separately:

1. Under $\varepsilon_n \leq \bar{\varepsilon}$, the first claim follows by the computation of the geometric series

$$\sum_{k=0}^{n-1} \rho^{n-1-k} \leq \sum_{k=0}^{\infty} \rho^k = \frac{1}{1 - \rho}.$$

2. Assuming that $\varepsilon_n \leq c_\varepsilon q^n$ for some $q \in (0, 1)$ and $c_\varepsilon > 0$ we have

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + c_\gamma c_\varepsilon \sum_{k=0}^{n-1} \rho^{n-1-k} q^k.$$

Suppose that $q > \rho$, then we have

$$\sum_{k=0}^{n-1} \rho^{n-1-k} q^k = \sum_{k=0}^{n-1} q^{n-1-k} \rho^k = q^{n-1} \sum_{k=0}^{n-1} (\rho/q)^k \leq q^{n-1} \frac{1}{1 - \rho/q} = q^n \frac{1}{q - \rho}$$

where we have used the formula for geometric series. Similarly, in the case of $q < \rho$, we can directly bound

$$\sum_{k=0}^{n-1} \rho^{n-1-k} q^k = \rho^{n-1} \sum_{k=0}^{n-1} (q/\rho)^k \leq \rho^{n-1} \frac{1}{1 - q/\rho} = \rho^n \frac{1}{(\rho - q)}.$$

Finally, in the case $q = \rho$, we simply bound $q^k < \tilde{q}^k$ for any $q < \tilde{q} < 1$ and deduce the claim line by line as before.

3. In the setting of $\varepsilon_n \leq \frac{c_\varepsilon}{(n+1)^p}$, we rewrite the upper bound on $\mathcal{F}(\mu_n)$ as

$$\begin{aligned}
\mathcal{F}(\mu_n) &\leq \rho^n \mathcal{F}(\mu_0) + c_\gamma \sum_{k=0}^{n-1} \rho^k \varepsilon_{n-1-k} \\
&= \rho^n \mathcal{F}(\mu_0) + c_\gamma \left(\sum_{k=0}^{\lfloor n/2 \rfloor} \rho^k \varepsilon_{n-1-k} + \sum_{k=\lfloor n/2 \rfloor + 1}^{n-1} \rho^k \varepsilon_{n-1-k} \right) \\
&\leq \rho^n \mathcal{F}(\mu_0) + \frac{2^p c_\gamma c_\varepsilon}{n^p} \frac{1}{1-\rho} + c_\gamma c_\varepsilon \frac{\rho^{\lfloor n/2 \rfloor}}{1-\rho}.
\end{aligned}$$

Here, we have used $\varepsilon_{n-1-k} \leq \frac{c_\varepsilon}{(n-k)^p} \leq \frac{2^p c_\varepsilon}{n^p}$ for all $k \leq \lfloor n/2 \rfloor$ and $\varepsilon_k \leq c_\varepsilon$ for all $k \in \mathbb{N}$.

□